Mixture & Ratio – Practice Questions (Q21–Q40)

All questions skipped — attempt before reviewing solutions.

Question 21

Pinky is standing in a queue at a ticket counter. Suppose the ratio of the number of persons standing ahead of Pinky to the number of persons standing behind her in the queue is 3 : 5. If the total number of persons in the queue is less than 300, then the maximum possible number of persons standing ahead of Pinky is:

(Type in the Answer — numerical answer required)

Question 22

A mixture P is formed by removing a certain amount of coffee from a coffee jar and replacing the same amount with cocoa powder. The same amount is again removed from mixture P and replaced with the same amount of cocoa powder to form a new mixture Q. If the ratio of coffee and cocoa in the mixture Q is 16 : 9, then the ratio of cocoa in mixture P to that in mixture Q is

A) 1:3
B) 1:2
C) 5:9
D) 4:9

Question 23

In an examination, Rama's score was one-twelfth of the sum of the scores of Mohan and Anjali. After a review, the score of each of them increased by 6. The revised scores of Anjali, Mohan, and Rama were in the ratio 11:10:3. Then Anjali's score exceeded Rama's score by

A) 26
B) 32
C) 35
D) 24

Question 24

Two alcohol solutions, A and B, are mixed in the proportion 1:3 by volume. The volume of the mixture is then doubled by adding solution A such that the resulting mixture has 72% alcohol. If solution A has 60% alcohol, then the percentage of alcohol in solution B is

A) 90%
B) 94%
C) 92%
D) 89%

Question 25

Bottle 1 contains a mixture of milk and water in 7:2 ratio and Bottle 2 contains a mixture of milk and water in 9:4 ratio. In what ratio of volumes should the liquids in Bottle 1 and Bottle 2 be combined to obtain a mixture of milk and water in 3:1 ratio?

A) 27:14
B) 27:13
C) 27:16
D) 27:18

Question 26

If a, b, c are three positive integers such that a and b are in the ratio 3:4 while b and c are in the ratio 2:1, then which one of the following is a possible value of (a + b + c)?

A) 201
B) 205
C) 207
D) 210

Question 27

Consider three mixtures — the first having water and liquid A in the ratio 1:2, the second having water and liquid B in the ratio 1:3, and the third having water and liquid C in the ratio 1:4. These three mixtures of A, B, and C, respectively, are further mixed in the proportion 4:3:2. Then the resulting mixture has

A) The same amount of water and liquid B
B) The same amount of liquids B and C
C) More water than liquid B
D) More water than liquid A

Question 28

A glass is filled with milk. Two-thirds of its content is poured out and replaced with water. If this process of pouring out two-thirds the content and replacing with water is repeated three more times, then the final ratio of milk to water in the glass, is

A) 1:27
B) 1:80
C) 1:81
D) 1:26

Question 29

A sum of money is split among Amal, Sunil and Mita so that the ratio of the shares of Amal and Sunil is 3:2, while the ratio of the shares of Sunil and Mita is 4:5. If the difference between the largest and the smallest of these three shares is Rs.400, then Sunil's share, in rupees, is

(Type in the Answer — numerical answer required)

Question 30

A fruit seller has a total of 187 fruits consisting of apples, mangoes and oranges. The number of apples and mangoes are in the ratio 5:2. After she sells 75 apples, 26 mangoes and half of the oranges, the ratio of number of unsold apples to number of unsold oranges becomes 3:2. The total number of unsold fruits is

(Type in the Answer — numerical answer required)

Question 31

A shop wants to sell a certain quantity (in kg) of grains. It sells half the quantity and an additional 3 kg of these grains to the first customer. Then, it sells half of the remaining quantity and an additional 3 kg of these grains to the second customer. Finally, when the shop sells half of the remaining quantity and an additional 3 kg of these grains to the third customer, there are no grains left. The initial quantity, in kg, of grains is

A) 50
B) 36
C) 42
D) 18

Question 32

Suppose C1, C2, C3, C4, and C5 are five companies. The profits made by C1, C2, and C3 are in the ratio 9:10:8 while the profits made by C2, C4, and C5 are in the ratio 18:19:20. If C5 has made a profit of Rs 19 crore more than C1, then the total profit (in Rs) made by all five companies is

A) 438 crore
B) 435 crore
C) 348 crore
D) 345 crore

Question 33

Two types of tea, A and B, are mixed and then sold at Rs. 40 per kg. The profit is 10% if A and B are mixed in the ratio 3:2, and 5% if this ratio is 2:3. The cost prices, per kg, of A and B are in the ratio

A) 17:25
B) 18:25
C) 19:24
D) 21:25

Question 34

A stall sells popcorn and chips in packets of three sizes: large, super, and jumbo. The numbers of large, super, and jumbo packets in its stock are in the ratio 7:17:16 for popcorn and 6:15:14 for chips. If the total number of popcorn packets in its stock is the same as that of chips packets, then the numbers of jumbo popcorn packets and jumbo chips packets are in the ratio

A) 1:1
B) 8:7
C) 4:3

D) 6:5

Question 35

Amala, Bina, and Gouri invest money in the ratio 3:4:5 in fixed deposits having respective annual interest rates in the ratio 6:5:4. What is their total interest income (in Rs) after a year, if Bina's interest income exceeds Amala's by Rs 250?

A) 6350
B) 6000
C) 7000
D) 7250

Question 36

The strength of a salt solution is p% if 100 ml of the solution contains p grams of salt. Each of three vessels A, B, C contains 500 ml of salt solution of strengths 10%, 22%, and 32%, respectively. Now, 100 ml of the solution in vessel A is transferred to vessel B. Then, 100 ml of the solution in vessel B is transferred to vessel C. Finally, 100 ml of the solution in vessel C is transferred to vessel A. The strength, in percentage, of the resulting solution in vessel A is

A) 15
B) 13
C) 12
D) 14

Question 37

Raju and Lalitha originally had marbles in the ratio 4:9. Then Lalitha gave some of her marbles to Raju. As a result, the ratio of the number of marbles with Raju to that with Lalitha became 5:6. What fraction of her original number of marbles was given by Lalitha to Raju?

A) 1/5
B) 6/19
C) 1/4
D) 7/33

Question 38

A trader sells 10 litres of a mixture of paints A and B, where the amount of B in the mixture does not exceed that of A. The cost of paint A per litre is Rs. 8 more than that of paint B. If the trader sells the entire mixture for Rs. 264 and makes a profit of 10%, then the highest possible cost of paint B, in Rs. per litre, is

A) 16
B) 26
C) 20
D) 22

Question 39

A chemist mixes two liquids 1 and 2. One litre of liquid 1 weighs 1 kg and one litre of liquid 2 weighs 800 gm. If half litre of the mixture weighs 480 gm, then the percentage of liquid 1 in the mixture, in terms of volume, is

A) 80

B) 70

C) 85

D) 75

Question 40

The salaries of Ramesh, Ganesh and Rajesh were in the ratio 6:5:7 in 2010, and in the ratio 3:4:3 in 2015. If Ramesh's salary increased by 25% during 2010-2015, then the percentage increase in Rajesh's salary during this period is closest to

A) 10

B) 7

C) 9

D) 8
